

Accelerated Preference Elicitation with LLM-Based Proxies

David Huang¹, Francisco Marmolejo-Cossío^{1,3,4}, Edwin Lock², and David Parkes¹

¹ Harvard University,
dhuang@college.harvard.edu, fjmarmol@g.harvard.edu, parkes@eecs.harvard.edu

² University of Oxford,
edwin.lock@cs.ox.ac.uk

³Boston College,

⁴IOG

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Abstract

Bidders in combinatorial auctions face significant challenges when describing their preferences to an auctioneer. Classical work on preference elicitation focuses on query-based techniques inspired from proper learning—often via *proxies* that interface between bidders and an auction mechanism—to incrementally learn bidder preferences as needed to compute efficient allocations. Although such elicitation mechanisms enjoy theoretical query efficiency, the amount of communication required may still be too cognitively taxing in practice.

We propose a family of efficient LLM-based proxy designs for eliciting preferences from bidders using natural language. Our proposed mechanism combines LLM pipelines and DNF-proper-learning techniques to quickly approximate preferences when communication is limited. To validate our approach, we create a testing sandbox for elicitation mechanisms that communicate in natural language. In our experiments, our most promising LLM proxy design reaches approximately efficient outcomes with five times fewer queries than classical proper learning based elicitation mechanisms.

Code — <https://github.com/davidzqhuang/alpha>.

1 Introduction

Auctions are fundamental to both theoretical and practical economics, enabling the efficient allocation of resources across a variety of domains, from spectrum sales Cramton [1997] to procurement Klemperer [2004] and beyond Palacios-Huerta et al. [2024]. Direct revelation mechanisms in combinatorial auctions, which require bidders to fully articulate their preferences, place impractical cognitive demands on bidders Budish and Kessler [2022], and larger auction instances quickly become prohibitive Nisan and Segal [2006], Braziunas [2006]. Work on preference elicitation aims to address these issues by letting bidders incrementally reveal their preferences. Instead of needing to produce the entirety of their preferences, bidders only need to produce comparatively simpler responses to queries, and only communicate enough information to clear the auction. In this context, the query complexity of elicitation mechanisms can be polynomial in the size of the description of a valuation Lahaie and Parkes [2004]. However, even a polynomial query complexity can be prohibitive in practice if bidders face significant cognitive and psychological costs to introspect and articulate their preferences in response to queries Ergin and Sarver [2010], Nielsen and Rehbeck [2022].

In response to these challenges, we introduce a novel approach of communicating preferences using natural language. We integrate LLMs into the framework of incremental revelation mechanisms, specifically the competitive equilibrium combinatorial auction (CECA) Lahaie and Parkes [2004]. In our approach, LLM-powered proxies act as intermediaries in the elicitation process, interacting with the auction mechanism on behalf of the bidder. The proxies query the bidder through natural language

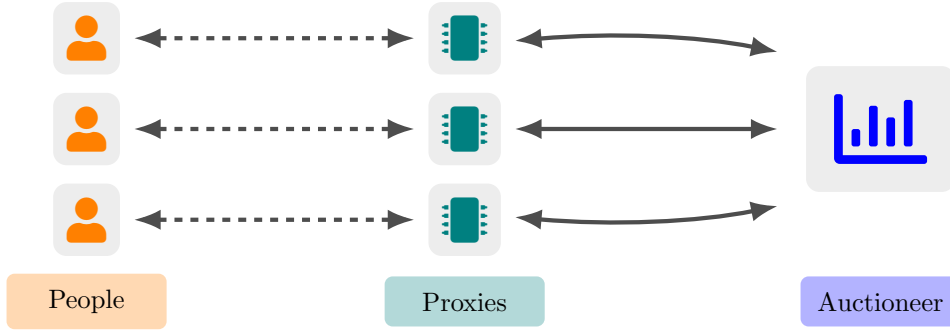


Figure 1: Overview of our proxy auction design with people, proxies and the auctioneer. The auctioneer communicates with the proxies to run the auction, and cannot communicate with the people directly. Each person’s proxy maintains a belief of its person’s preferences, which it refines over time by communicating with its person using value queries, demand queries and natural language questions.

interactions to uncover preferences. The use of LLMs allows the proxy to infer bidder preferences more efficiently than traditional methods.

To experimentally test the effectiveness of proxies, we develop a comprehensive pipeline that simulates the bidding process across multiple items. The pipeline makes an additional use of LLMs to simulate human bidder behavior (cf. Horton [2023]) when interacting with a language-based proxy. In our approach, bidder “types” are dictated by a seed description in natural language that encapsulates their preferences, and a *simulation LLM* (distinct from the LLM-powered proxy) uses this seed to generate responses to queries. We show that our seed descriptions are robust by demonstrating that different LLMs give similar valuations for the same bundle when shown the same seed. In our experiments, the LLM proxy interacts with these simulation LLMs, querying them to learn about their underlying preferences.

After the auction concludes, we evaluate the social welfare by summing the values each simulated bidder ascribes to their allocated bundle and comparing this to the welfare-maximizing outcome given full knowledge of the simulated bidders’ implied valuations. This approach allows us to measure not only the accuracy of preference elicitation but also the allocate efficiency of LLMs serving as proxies.

We benchmark the performance of the LLM proxy against proxies that make use of proper learning methods to infer LLM bidder preferences Lahaie and Parkes [2004]. We vary the number of queries they are permitted to make to bidders, assessing both how well they learn bidder preferences and the overall efficiency of the eventual allocation. The results are encouraging: our most promising LLM proxy design demonstrates the ability to elicit preferences that allow the auction to compute outcomes with a higher welfare using five times fewer queries than traditional elicitation methods. Our findings suggest that integrating LLMs into auction mechanisms presents a promising avenue for enhancing the practical deployment of combinatorial auctions.

1.1 Related Literature

Recent work has significantly advanced preference elicitation and efficiency in combinatorial auctions by leveraging machine learning techniques. Approaches like *probably approximately efficient auctions* Brero et al. [2017] and *ML-powered iterative combinatorial auctions* Brero et al. [2019b] use bidder-specific models to generalize valuations and optimize allocations, while neural network architectures such as *monotone-value neural networks (MVNNs)* Weissteiner et al. [2021] and deep learning frameworks Weissteiner and Seuken [2020], Soumalias et al. [2024] improve scalability. Bayesian frameworks Brero et al. [2019a], Weissteiner et al. [2023] and multi-task learning techniques Maruo and Kashima [2024], Soumalias et al. [2024] further enhance elicitation by modeling uncertainty and sharing parameters across bidders. However, these methods rely on structured query responses and lack natural language integration. Our work introduces a fundamentally different approach by leveraging large language models (LLMs) as proxies for elicitation, requiring natural language descriptions of auction instances—a capability absent from current benchmarks such as SATS Weiss et al. [2017]. Unlike prior studies that validate LLMs in narrow domains Austin et al. [2024], Ronanki et al. [2023], we integrate LLMs with proper learning techniques to approximate preferences through natural language, reducing bidders’ cognitive load and limiting com-

munication overhead. This novel methodology applies LLMs as elicitation methods to combinatorial auction settings, representing a complementary direction in the field.

2 The Proxy Auction Model

We consider a *proxy auction design* with an auctioneer selling goods $G := \{g_1, \dots, g_n\}$, and a population of people $P := \{p_1, \dots, p_m\}$ who each wish to acquire a subset, or *bundle*, of G . The auctioneer and people do not communicate with each other directly. Instead, each person $p \in P$ is associated with a (distinct) proxy $\omega(p)$, who maintains a belief of the person’s preferences and interacts with the auctioneer on her behalf. Figure 1 illustrates this setup.

Let $\mathcal{B} := \{b \mid b \subseteq G\}$ denote the collection of all bundles. A *XOR bid* $\theta = (B, v)$ consists of a set B of *atomic bundles* satisfying $\{\emptyset\} \subseteq B \subseteq \mathcal{B}$ and *atomic valuation* $v : B \rightarrow \mathbb{R}_+$. Bid θ induces the valuation v^θ that is defined, for every bundle $b \in \mathcal{B}$, by

$$v^\theta(b) := \max_{\{b' \in B \mid b' \subseteq b\}} v(b'). \quad (1)$$

So the value of a bundle is determined by the highest-valued atomic bundle it contains. XOR bids can represent any valuation v satisfying the *free disposal* property that $v(b) \leq v(b')$ if $b \subseteq b'$.

The auctioneer sets prices defined by a *price function* $\varphi^i : \mathcal{B} \rightarrow \mathbb{R}_+$ for each person $p_i \in P$. Prices can be non-linear and non-anonymous, so the price assigned a bundle is not necessarily the sum of the prices of its items, and different buyers may face different prices for the same bundle.

The *demand correspondence* D^θ of XOR bid θ maps every price function φ to the set of bundles that maximize quasi-linear utility at these prices, so

$$D^\theta(\varphi) := \arg \max_{b \in \mathcal{B}} v^\theta(b) - \varphi(b).$$

A *competitive equilibrium* for XOR bids $\theta^1, \dots, \theta^m$ consists of a price function φ^i and allocation b^i for each bid θ^i such that $b^i \in D^{\theta^i}(\varphi^i)$ and $(b^1, \dots, b^m)$ partitions the set of goods G . Bikhchandani and Ostroy [2002] show that a competitive equilibrium is guaranteed to exist for any collection of XOR bids, and maximizes *social welfare* $\sum_{i \in [m]} v^{\theta^i}(b^i)$ among all partitions $(b^1, \dots, b^m)$ of the goods. The goal of the auctioneer is to find a welfare-maximizing outcome, which he does by computing a competitive equilibrium.

Communication between proxy and person. Each proxy maintains a candidate XOR bid, which represents its current belief of its person’s preferences. Each proxy gathers information about its person’s preferences by making value queries, demand queries, or natural language queries to the person. We write $\mathcal{Q}_N^p(q)$ for the answer from person p to natural language query q . Similarly, $\mathcal{Q}_V^p$ maps any bundle to a (non-negative real) value that person p has for the bundle. Finally, a demand query $\mathcal{Q}_D^p$ takes prices φ and a bundle b as input and returns either $(1, b)$ if the person is happy with b at φ or $(0, b')$ if the person would prefer some bundle $b' \neq b$ to b at φ . Answers to queries need not be deterministic. Sections 3 and 4 outlines different variations on proxy designs.

Each proxy, $\omega(p)$, maintains a *context* that it updates by repeatedly communicating with p . The proxy uses this context to formulate its beliefs of p ’s preferences. In our LLM-powered proxies, this context includes the *communication history* of queries and responses stored in natural language.¹

Natural language descriptions. An integral part of our setup is that a given auction instance, $\mathcal{A}$, is associated with a *description function* that maps $\mathcal{A}$ to a natural language description of the auction itself, its people, and its goods. For example, auction $\mathcal{A}$ might be described by $\mathcal{D}(\mathcal{A}) = \text{“A farmer’s market. There are three goods: an apple, a banana, and a strawberry. We have one person, Carlos, who only likes red fruit and has a budget of \$7.”}$. The description function of an auction will be used by our LLM proxy designs, as well as in our LLM pipeline to simulate people in our experiments.

¹In our software implementation, the LLM proxies maintain a history containing a transcript of all communications in natural language. Our benchmark DNF learning proxy simply maintains a candidate XOR bid as its context.

2.1 The Competitive Equilibrium Combinatorial Auction

The auctioneer implements the *competitive equilibrium combinatorial auction (CECA)* Lahaie and Parkes [2004] in our proxy auction setting. In this auction, the auctioneer first asks each proxy for a XOR bid, and then computes a competitive equilibrium for these bids using integer linear programming. The auction then asks the proxies of people not satisfied with their allocation at their prices to update their XOR bids, and repeats the procedure. The proxies thus incrementally elicit new information until a competitive equilibrium satisfying all people can be reached.

Given a XOR bid θ , we define prices φ^θ as $\varphi^\theta(b) := v^\theta(b)$ for each $b \in \mathcal{B}$. These prices support any competitive equilibrium allocation Lahaie and Parkes [2004].² We can now formulate the CECA in detail:

- Step 1.** Retrieve a XOR bid θ^i from each proxy $\omega(p_i)$, and compute a competitive equilibrium $(\varphi^{\theta^i}, b^i)$ with maximal competitive equilibrium prices for these bids using integer linear programming.
- Step 2.** Ask every proxy whether their person is satisfied with the allocation b^i at prices φ^{θ^i} . If so, **return** the competitive equilibrium.
- Step 3.** Request the proxy of every unsatisfied person to update their XOR bid, and go to Step 1.

2.2 Proxy Implementations

The CECA is straightforward to implement if all proxies, ω , implement a *message subroutine*, $\mathcal{M}^\omega$, that the auctioneer uses to communicate with the proxies. The subroutine takes as input a bundle b and prices φ . If the person p represented by $\omega = \omega(p)$ is satisfied with bundle b at prices φ , the subroutine returns $(1, \theta)$, where θ is the proxy's current candidate XOR bid. Otherwise, the subroutine returns $(0, \theta')$, where θ' is an updated candidate XOR bid incorporating the new information that the person is unsatisfied with b at φ . When participants satisfy quasilinear preferences, the proxy implementation of the CECA auction is guaranteed to terminate with the efficient allocation as long as new information is provided in the updated bid θ' Lahaie and Parkes [2004] (see also Section 3).

3 The DNF Learning Proxy

We first describe the *DNF learning proxy*, denoted ω_{XOR} , that is adapted from the preference elicitation literature Lahaie and Parkes [2004]. Fix a person p . The proxy ω_{XOR} maintains a candidate XOR bid $\theta = (B, v)$, initialized as $B = \{\emptyset\}$ and $v(\emptyset) = 0$. It implements the message subroutine $\mathcal{M}^{\omega_{\text{XOR}}}$ from Section 2.2 as the following procedure.

Given a bundle b and prices φ as input, it makes a demand query $Q_D^p(\varphi, b)$ to its person, p , and considers the query's outcome (r, b') . If $r = 1$, then the person is satisfied with b at φ , so the algorithm returns $(1, \theta)$. Otherwise, if $r = 0$, the algorithm finds a new atomic bundle b'' and value $v(b'')$ to add to the XOR bid θ that it maintains. It then returns $(0, \theta')$, with θ' being the updated candidate XOR bid.

When searching for a new atomic bundle, b'' , the proxy queries and stores its person's value, x , of the initial bundle b' . It then iterates over all items, i , in the bundle (e.g., in ascending order) in turn, removing the item from the current bundle b' whenever the value of b' is equal to the value of the bundle b' without item i . Following this loop, the resulting bundle is then returned as b'' , together with its value, x .

This procedure is derived from existing proper learning algorithms for monotone DNF formulas Blum et al. [2004]. It is immediate that the new atomic bundle b' and its value can be found with at most n valuation queries. With quasilinear preferences, the proxy will correctly learn the person's valuation in the form of $\theta = (B, v)$ after at most $|B|$ demand queries and $|B|n$ valuation queries. See Lahaie and Parkes [2004] for more detail.

In our experiments, the DNF learning proxy serves as the benchmark from the preference elicitation literature against which we compare our LLM-based proxies.

²Although competitive equilibrium allocations are supported by a lattice of prices, we follow Lahaie and Parkes [2004] in describing the CECA with maximal competitive equilibrium prices.

4 Our LLM Proxy Designs

We now propose a family of proxy designs that incrementally build upon the DNF learning proxy by incorporating LLMs. As mentioned previously, all our LLM proxies maintain a history of past interactions with their person and the auctioneer in the form of a natural language transcript. This transcript contains all queries made to its person and as well as her answers, and all message requests from the auctioneer and the XOR bids returned by the proxy in response. Additionally, the proxies maintain a belief of their person’s preference in the form of a XOR bid $\theta = (B, v)$, which we call the ‘candidate XOR bid’. The proxy uses their transcript and candidate XOR bid to implement the subroutine $\mathcal{M}^\omega$ from Section 2.1.

Throughout this section, we fix a person $p \in P$ for which we describe the proxy, shortening $\omega(p)$ to ω for convenience.

4.1 Drop-in LLM Proxies

Drop-in LLM proxies are ‘drop-in’ replacements for proper-learning proxies like ω_{XOR} because they communicate with their person using the same message types: value queries and demand queries. We propose two such designs which incrementally build upon the baseline DNF learning proxy.

Design 1. The LLM proxy ω_{VD1} implements the message subroutine $\mathcal{M}^{\omega_{\text{VD1}}}$ as the following procedure:

First it prompts an LLM with the input prices φ and allocation b , as well as the auction description $D(\mathcal{A})$ and the proxy’s transcript. Then it ask the LLM to choose one of the following actions:

1. indicate that the person is satisfied;
2. message the person a value query;
3. message the person a demand query.

If the LLM chooses action 1, it returns $(1, \theta)$, where θ is proxy’s current candidate XOR bid.

For action 2, the LLM is also asked to provide a bundle b with which to augment the proxy’s candidate XOR bid. The proxy then makes the query $\mathcal{Q}_V^p(b)$ to its person p to ascertain her value, z , for bundle b , and extends $\theta = (B, v)$ to a new XOR bid θ' by adding b to B and letting $v(b) = z$. Subsequently, the proxy returns $(1, \theta')$.

For action 3, the proxy make a demand query $\mathcal{Q}_D^p(\varphi, b)$ and retrieves a bundle b' demanded by p . If $b' = b$, then it returns $(0, \theta)$. Otherwise, it makes the query $\mathcal{Q}_V^p(b')$ to ascertain person p ’s value, x for b' and returns $(1, \theta')$, where θ' is the augmented XOR bid as above.

Design 2. The second drop-in proxy ω_{VD2} is similar to ω_{VD1} . As with ω_{VD1} , it maintains an internal candidate XOR bid, $\theta' = (B', v')$ as a part of its context. Unlike ω_{VD1} , the proxy ω_{VD2} reports to the auctioneer an *inferred* XOR bid $\theta^* = (B^*, v^*)$, which agrees with θ' for all bundles $b \in B'$, and additionally infers values $v^*(b)$ for all $b \in \mathcal{B} \setminus B'$.

ω_{VD2} computes θ^* using an inference function $\gamma : \mathcal{B} \rightarrow \mathbb{R}_+$ that gives an inferred value for each bundle. To do this, the proxy follows the following procedure: it maintains a working $\theta^* = (B^*, v^*)$ initialized to θ' , and iterates over all possible bundles in ascending order of the number of items they contain and for each bundle b , if $b \notin B'$ and $v^*(b) < \gamma(b)$, it adds b to B^* and defines $v^*(b) = \gamma(b)$.

The inference function γ is calculated at scheduled iterations of the CECA auction, including the first iteration, using an LLM pipeline. The LLM, given the proxy’s transcript, the auction description $D(\mathcal{A})$, and a bundle b , infers a value z for the bundle. Thus γ can be constructed by calling the pipeline with all $b \in \mathcal{B} \setminus B'$. Finally, ω_{VD2} discounts the value given by the LLM by a constant factor of $\epsilon = 0.75$ to prevent ω_{VD2} from over-reporting its person’s valuation for bundles. The reason for this is two-fold: 1) over-estimation of valuation can lead to artificially high prices for bundles which a person could be allocated otherwise, and 2) artificially high prices can result in a person over-paying and obtaining negative utility from participating in the auction.

4.2 Plus-Questions LLM Proxy

The *plus-questions proxy* ω_{NVD} uses natural language questions in addition to value queries and demand queries when messaging their person. The proxy implements $\mathcal{M}^{\omega_{\text{NVD}}}$ almost identically to the implementation of ω_{VD2} , except that the very first time it is run, it first calls $\mathcal{Q}_N^p(q)$, where question q is given by

the LLM response to a prompt asking, “What should the proxy ask the person next to better understand their preferences?” given the auction description $\mathcal{D}(\mathcal{A})$.

Note that ω_{NVD} , like ω_{VD2} , also returns an inferred XOR bid θ^* . The additional information that person p provides in response to q is recorded in the proxy’s transcript, and thus is used in all subsequent calls to the message subroutine to give more information to the LLM when it chooses which action to take, and how to infer its new candidate XOR bid.

4.3 Hybrid Proxy

The *hybrid proxy* ω_{H} combines ω_{NVD} and the ω_{XOR} . The procedure implementing $\mathcal{M}^{\omega_{\text{H}}}$ has two parameters, $\alpha \in \mathbb{Z}_+$ and $\delta \in (0, 1)$.

For the first α calls to $\mathcal{M}^{\omega_{\text{H}}}$, the procedure simulates the procedure of ω_{NVD} that implements $\mathcal{M}^{\omega_{\text{NVD}}}$. So the proxy initially asks a natural language question and uses its LLM to infer valuations. After α iterations, the procedure instead simulates the procedure of ω_{XOR} implementing $\mathcal{M}^{\omega_{\text{XOR}}}$ and thus uses the proper-learning approach from the preference elicitation literature. However, recall how $\mathcal{M}^{\omega_{\text{VD2}}}$ periodically recomputes the inference function γ and, on each call, calculates and return the inferred valuation θ^* . The procedure of ω_{H} continues to derive an inferred valuation after simulating the procedure of ω_{XOR} . We specify that in every iteration after the switchover, we update γ to $\gamma' = \delta\gamma$. Multiplying with δ , which lies in $(0, 1)$, has the purpose of maintaining, in the short run, the benefits of using an inferred valuation, but in the long run, removing the inferred valuation by decaying γ towards 0.

5 The LLM Simulation Pipeline

In order to evaluate the effectiveness of the DNF learning proxy and our LLM proxies, we make another use of LLMs, also designing an LLM pipeline for simulating people in our auction setup. This is used in Section 6 for our experiments. In our simulations, the LLM answer natural language questions as well as value and demand queries. The validity of our experimental results are subject to the validity of the LLM being a reasonable model for how people will respond to natural language queries, as investigated in Brand et al. [2023], Horton [2023].

Preferences are developed by our simulation pipeline by constructing a *seed* description, $\mathcal{D}(p)$, in natural language for each simulated person p . These seeds are crucial for the simulated person to be able to answer natural language questions about their preferences. This contrasts with other methods of generating random preferences, e.g., simulated preferences that arise from arbitrary orderings of bundles. Such methods do not necessarily represent realistic preferences and, secondly, cannot be described meaningfully in natural language.

5.1 Seed Generation LLM Pipeline

The process for generating the seeds, $\mathcal{D}(p)$, that define person p ’s preferences consists of a four-step LLM pipeline. The first step generates the germ of the seed, and the three subsequent steps revises the seed to be more specific and more coherent.

Step 1: Initial preference generation. The LLM generates a person’s specific purchase occasion and taste over the items, providing an initial description of a person’s preferences based on the given auction scenario. To encourage diversity in responses, we ask the LLM to generate three possible versions and then randomly sample one.

Step 2: Revision for clarity and consistency. The LLM revises the initial preferences to enhance clarity and internal consistency in describing the person’s preferences.

Step 3: Anchor with specific values. The LLM anchors the person’s preferences by assigning explicit dollar values to specific bundles of interest.

Step 4: Refinement for precision in complex bundles. The LLM refines the person’s preferences to outline a process that defines how they evaluate other unspecified bundles, considering items as substitutes and complements.

	Electronics	Preserves	Transportation
Description	A collection of like-new electronics donated to a local library, ranging from AirPods to iPads and accessories.	Gourmet food items are available, including exotic jams and chutneys with unique flavors, perfect for cooking or gourmet meals.	A variety of electric scooters and bikes are up for auction, catering to urban commuters and fitness enthusiasts.
Items	Apple AirPods (2nd Gen) Apple AirPods Max Apple iPad (9th Gen) Apple iPad Air (M2) Apple Pencil (2nd Gen) Apple Pencil Pro	Organic Strawberry Jam Wild Blueberry Preserves Apricot & Lavender Conserve Sugar-Free Raspberry Spread Spiced Plum Chutney Mango & Passionfruit Jam	Electric Scooter S2 Pro (2024) E-Scooter Elettrica (2023) Voltron SP03 E-Scooter (2024) Troik Verve+ 2 (2023) Titan Escape 3 (2023) Schwin Suburban (2021)

Table 1: The three auction descriptions used in our experiments.

5.2 Message Response LLM-Based Pipeline

In order to ensure that the simulated person acts coherently, consistent with a standard model of preferences, the pipeline initially constructs an XOR bid $\theta = (B, v)$ for the person, which is then cached. The pipeline then responds to value and demand queries using θ .

We implement a subroutine to perform LLM calls and help construct θ . This subroutine takes as input a bundle and returns an *initial* value for this bundle. Maintaining a working valuation $\theta = (B, v)$, we iterate over all possible bundles in ascending order of the number of items they contain. At each iteration, we run the first subroutine to get the *initial* value ν of the bundle and update θ with that bundle and its value if free disposal is not violated, i.e, if $v^*(b) < \nu$.

6 Experiments

The experiments, along with the simulation toolbox, were implemented in Python 3. We used Google’s OR-Tools package for integer linear programming for the auction. OpenAI’s gpt-4o-mini model (versioned to July 18, 2024) is used as our LLM for the proxy and the simulation pipeline.

The three auction scenarios we use in our experimental evaluation of our LLM proxies are outlined in Table 1. They are designed to be accessible so that the reader can readily assess the realism of the interactions and various LLMs can be tested without domain-specific knowledge or fine tuning. Each scenario has $N = 6$ items for auction. Some items are clear substitutes and others are clear complements. For each scenario, we generated three tranches of three seeds each, for a total of 27 simulated participants in the auctions.

In Sections 6.3 and 6.4, we run experiments on the LLM simulation pipeline to validate the robustness and sensibility of the preferences of the simulated people created by the pipeline. In Sections 6.1 and 6.2, we run experiments to test the performance of our LLM proxy designs against the benchmark DNF learning proxy design.

6.1 Auction Efficiency

We explore a set of simulations of auctions to see how quickly LLM proxies (drop-in and plus-questions proxies) reach efficient outcomes, compared to the DNF learning proxy. Our simulations show that efficient outcomes can be reached with less communication using LLM-based proxies. Quantitatively, we can reach 75% efficiency in ten interactions person-proxy interactions with ω_{XOR} , in four interactions with ω_{VD2} , and in two interactions with ω_{NVD} (see Fig. 2).

For the nine tranches of three people each with three tranches per scenario, we run the auction using either ω_{XOR} , ω_{VD1} , ω_{VD2} , or ω_{NVD} for all people, halting the simulation after 16 iterations. We fully elicit each of the 27 people’s preferences across all 2^6 bundles, and use that to calculate the welfare-maximizing outcome.

Each iteration of the competitive equilibrium auction, we calculate the social welfare of the auction and record the number of interactions between each person and their proxy. We then calculate, for each iteration, the percent efficiency by dividing the social welfare by that of the social welfare maximizing outcome. We also calculate the average number of interactions. We then determine for $i = 1, 2, \dots, 16$

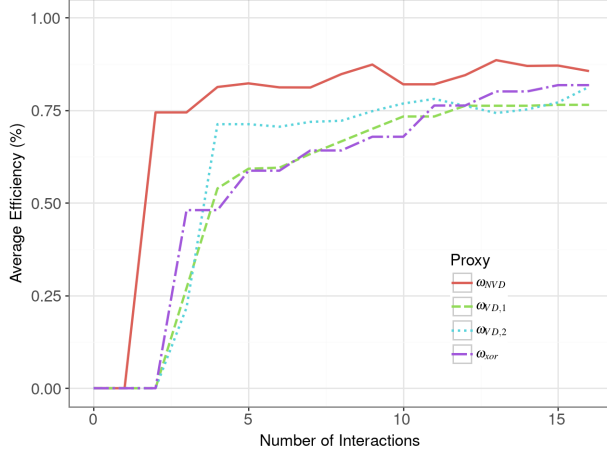


Figure 2: Average efficiency of the auction run under the three scenarios from Table 1. Anchoring on our DNF learning proxy, we see that the drop-in LLM proxy ω_{VD1} performs similarly, also reaching 75% efficiency in around 10 person-proxy interactions. The proxy ω_{VD2} performs significantly better, reaching 75% efficiency in four interactions; proxy ω_{NVD} , which additionally uses a natural language question, reaches 75% efficiency in two interactions.

the efficiency with i interactions by selecting the efficiency at the iteration of the run that has the lowest number of interactions greater than i . Then, for each i , we average over all nine tranches to get the percent average efficiency shown in Fig. 2.

6.2 Long-Run Performance of Hybrid Proxy

Section 6.1 shows how LLM-based proxies outpace the proper-learning-based ω_{XOR} in the short run. We now consider the hybrid proxy ω_H to characterize the long-term performance of an LLM-based proxy, showing that we can combine the short-term rapid increase in efficiency without sacrificing the long-term ability to exactly learn the person’s valuation and to converge to the social-welfare-maximizing auction outcome. For the following experiments, we set the hybrid proxy parameters to $\alpha = 10$ and $\delta = 0.95$.

First, we run ω_H and ω_{XOR} as preference learning algorithms, by running the auction with only one person with either ω_H or ω_{XOR} as a proxy, constantly setting prices equal to the candidate XOR bid values and allocating the empty bundle, and measure at each step the number of times the proxy interacts with the person and the average error, calculated as the average absolute difference between the value of an item calculated using the person’s XOR bid and the value of an item calculated using the proxy’s candidate XOR bid. These numbers are aggregated over the multiple tranches using the methodology in Section 6.1.

The results, plotted in Fig. 3 (a), show that ω_H consistently approximates the person’s valuation with fewer interactions. In all three scenarios, ω_H achieves a better approximation of the person’s valuation than ω_{XOR} does in 50 interactions. We note, however, that at the very end, ω_{XOR} obtains 0 error, so it exactly learns the person’s valuation with approximately 10 fewer iterations. This is because the LLM is sometimes inefficient with its queries, and natural language questions are more helpful for approximations than exact learning.

Second, we ran the auction simulations identical to those in Section 6.1, except that we let the auction continue running until completion. We aggregate and plot the data according to the same procedure and plot the results in Fig. 3 (b). In the initial phase, ω_H rapidly reaches fairly efficient outcomes. This matches the results shown for ω_{NVD} . This is expected, as $\mathcal{M}_C^{\omega_H}$ simulates $\mathcal{M}_C^{\omega_{NVD}}$ in the first α iterations. We also see that the efficiency of the outcome steadily increases as we increase the number of interactions, in line with ω_{XOR} , so we reach an outcome that maximizes social welfare.

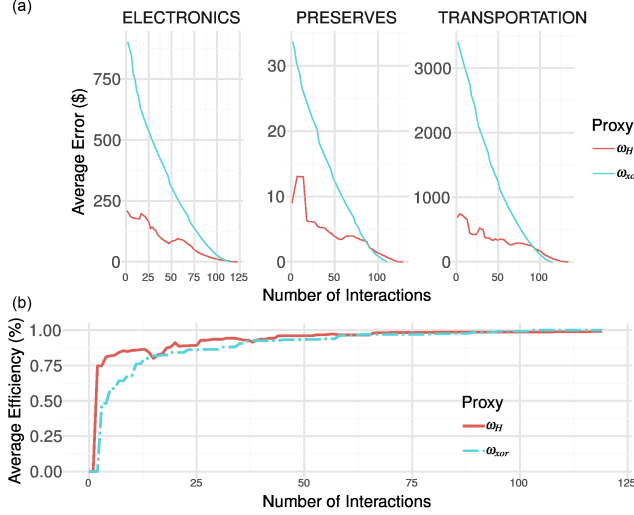


Figure 3: Performance of the hybrid proxy ω_H and the DNF learning proxy ω_{XOR} with respect to number of person-proxy interactions. Panel (a) shows that ω_H achieves a good approximation of its person’s valuation significantly quicker than ω_{XOR} , while still converging to the person’s exact valuation. Panel (b) shows a rapid efficiency increase initially for ω_H , and similar long-run efficiency.

6.3 Robustness of LLM-Based Person Modeling

We find that the seeds given by the seed-generation LLM pipeline of the simulation are robust. To understand whether the seeds robustly describe a specific valuation, we examine the inter-model robustness of the seeds, comparing the valuations generated by the message response LLM-pipeline using the gpt-4o-mini model to those generated with the gemini-1.5-flash model [DeepMind \[2024\]](#), we find that the valuations are broadly consistent, as shown in Fig. 4. As the models are developed independently by OpenAI and Google, respectively, this shows that the seeds, and the valuations interpreted from them, via the message-response pipeline are not specific to a specific family of models and thus robust.

6.4 Coherent Preferences

We find that the preferences created by the LLM simulation pipeline are coherent. For this, we investigate the preferences generated by the pipeline by running the following experiment. For a specific seed description in each scenario, start with the empty bundle and first incrementally add items until the bundle contains all items, before incrementally removing items again until the bundle is empty. After each addition and removal, we run the message response LLM pipeline to perform a value query, repeating this ten times to account for variability arising from randomness. We find that the valuations given are precise. The variation for a given bundle is tight and the overall shape satisfies desirable properties such as monotonicity and free disposal. Fig. 5 shows that the value of the bundle consistently increases as we add items to the bundle, and decreases as we remove items.

7 Conclusion

This study presents a novel approach to preference elicitation in combinatorial auctions by integrating Large Language Models (LLMs) into proxies. Traditional query-based elicitation methods, while theoretically efficient, often impose substantial cognitive burdens on people. Our LLM-powered proxy design mitigates this through natural language interactions and inference, leading to more human-centric and efficient auction mechanisms in increasingly complex markets.

Our experiments demonstrate that LLM-based proxies, particularly those that use LLMs to infer the person’s valuation on unspecified bundles, achieve approximately efficient auction outcomes up to five times faster than classical DNF learning proxies. As illustrated in Fig. 2, proxies such as ω_{VD2} and ω_{NVD}

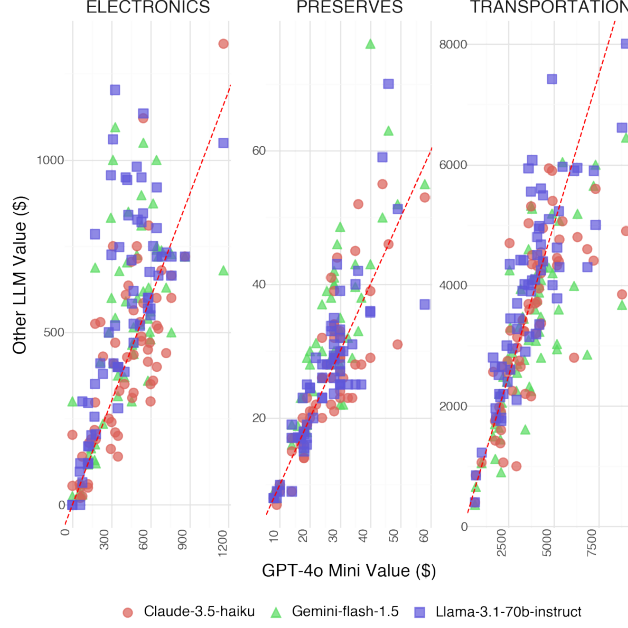


Figure 4: Comparison between GPT-based and other LLM models in person robustness tests. Gpt-4o-mini and other LLM-models give similar values in response to value queries evidenced by the dots, each of which representing a single bundle, falling near the identity gpt-4o-mini value (\$) = Other LLM value (\$) line.

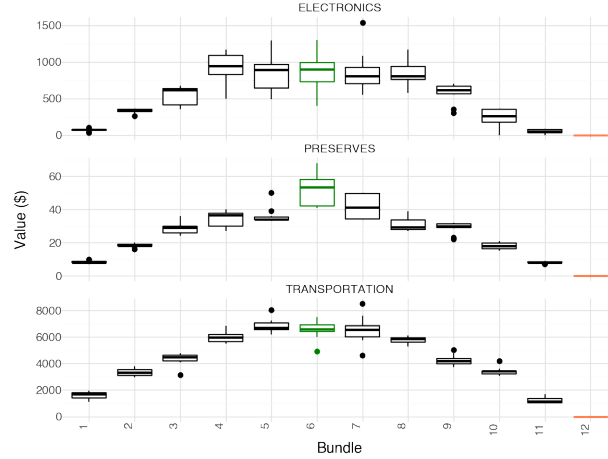


Figure 5: Preference variability and shape after iteratively adding items to an initially empty bundle, and then iteratively removing items. Each box plot displays the distribution of values obtained from with 10 iterations of the process.

reach 75% of maximum social welfare with markedly fewer interactions compared to the ω_{XOR} proxy. Furthermore, the hybrid proxy ω_{H} not only accelerates the initial elicitation phase but also converges to the social-welfare-maximizing outcome, suggesting a path forward for combining the rapid approximation capabilities of LLMs with the guarantees of classical preference elicitation.

Additionally, we validate the robustness of LLM-based person modeling across different models and repeated trials, ensuring consistent and reliable preference simulation. This robustness underscores the feasibility of employing natural language processing in complex auction settings, enhancing both the theoretical and practical aspects of auction design.

In summary, our findings highlight the potential of LLMs to improve preference elicitation in combinatorial auctions by making the process more intuitive and less burdensome by using natural language

questions and reducing the number of interactions for agents. In addition, we contribute a comprehensive testing sandbox for simulating bidders in auction environments that use natural language both to describe the auction and to communicate between auction participants and proxies.

In future work, the practicality and efficacy of our LLM proxies can be explored in lab experiments featuring human participants representing bidders instead of our LLM-powered simulation pipeline. In addition, further research could also explore the scalability of this approach to larger auction environments and investigate the integration of increasingly advanced LLM capabilities to further enhance elicitation efficiency and accuracy at lower cost.

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